

Quiz 2 – Solutions

ECON 308 – International Economic Relations

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Answers to Multiple-Choice Questions

1. **B) Capital owners.**

When P_M/P_F rises, labor moves toward M . The value marginal product of K rises both because of the higher P_M and because L_M increases. Landowners lose; workers are ambiguous.

2. **C) More labor moves into manufactures.**

Equilibrium requires $w = P_M \cdot MPL_M(L_M) = P_F \cdot MPL_F(L_F)$. A higher P_M shifts up the M -schedule, so equality holds only with larger L_M (and smaller L_F).

3. **C) Because real wage rises in terms of one good and falls in terms of the other.**

Typically w rises less than P_M , so w/P_M falls, but w/P_F rises when P_F is unchanged.

4. **C) Specific factors in the export sector gain, specific factors in the import sector lose, labor is ambiguous.**

Classic Ch. 4 distributional result: the specific factor in the sector whose relative price rises gains; the other specific factor loses; labor's real income is mixed.

5. **B) Home's opportunity cost of manufactures is lower than Foreign's.**

Home exports M under free trade only if $OC_M^{Home} < OC_M^{Foreign}$ (equivalently, P_M/P_F under autarky is lower at Home).

Solution to the Problem

Two goods, M and F . Labor L mobile across sectors; K specific to M ; T specific to F .
Production:

$$Q_M = K^{1/3} L_M^{2/3}, \quad Q_F = T^{1/2} L_F^{1/2}, \quad L_M + L_F = L.$$

Endowments: $K = 64$, $T = 100$, $L = 90$.

Prices: initially $P_M = 2$, $P_F = 1$. Later P_M rises to 3 (with $P_F = 1$).

1. **Marginal products of labor.**

$$MPL_M \equiv \frac{\partial Q_M}{\partial L_M} = \frac{2}{3} K^{1/3} L_M^{-1/3}, \quad MPL_F \equiv \frac{\partial Q_F}{\partial L_F} = \frac{1}{2} T^{1/2} L_F^{-1/2}.$$

Both are decreasing in own labor, so moving labor into a sector lowers its marginal product.

2. Initial equilibrium (L_M, L_F, w) **with** $P_M = 2, P_F = 1$.

Wage equalization (labor mobility):

$$w = P_M \cdot MPL_M(L_M) = P_F \cdot MPL_F(L_F), \quad L_F = 90 - L_M.$$

Plug constants $K^{1/3} = 4, T^{1/2} = 10$:

$$2 \cdot \frac{2}{3} \cdot 4 \cdot L_M^{-1/3} = 1 \cdot \frac{1}{2} \cdot 10 \cdot (90 - L_M)^{-1/2}.$$

Simplify:

$$\frac{16}{3} L_M^{-1/3} = 5 (90 - L_M)^{-1/2}.$$

This equation is monotone in L_M and has a unique interior solution. Solving numerically:

$$\boxed{L_M \approx 74.446, \quad L_F \approx 15.554}.$$

The common wage is (either side of the equality):

$$w = P_M \cdot MPL_M(L_M) = \frac{16}{3} L_M^{-1/3} \Rightarrow \boxed{w \approx 1.268}.$$

3. Outputs and specific-factor returns (Q_M, Q_F, r_K, r_T).

Outputs:

$$Q_M = K^{1/3} L_M^{2/3} = 4 L_M^{2/3} \Rightarrow \boxed{Q_M \approx 70.787},$$

$$Q_F = T^{1/2} L_F^{1/2} = 10 \sqrt{L_F} \Rightarrow \boxed{Q_F \approx 39.438}.$$

Specific-factor returns:

$$r_K = P_M \cdot \frac{\partial Q_M}{\partial K} = 2 \cdot \left(\frac{1}{3} K^{-2/3} L_M^{2/3} \right).$$

Since $K^{2/3} = 16, K^{-2/3} = 1/16$, so

$$r_K = 2 \cdot \frac{1}{3} \cdot \frac{1}{16} L_M^{2/3} = \frac{1}{24} L_M^{2/3} \Rightarrow \boxed{r_K \approx 0.737}.$$

For land:

$$r_T = P_F \cdot \frac{\partial Q_F}{\partial T} = 1 \cdot \left(\frac{1}{2} T^{-1/2} L_F^{1/2} \right) = \frac{1}{20} \sqrt{L_F} \Rightarrow \boxed{r_T \approx 0.197}.$$

With these endowments and prices, most labor goes to M . The high L_M raises M 's marginal productivity of K , so $r_K > r_T$.

4. Effects of a rise in P_M from 2 to 3 (qualitative, then numeric check).

Comparative statics (signs):

$$w = P_M \cdot MPL_M(L_M) = P_F \cdot MPL_F(L_F).$$

When P_M increases:

- The M -side wage schedule shifts up $\Rightarrow L_M$ **rises**, L_F falls.
- Because L_M rises, MPL_M falls, so w **rises less than proportionally** to P_M .
- r_K **rises more than proportionally**: it scales with P_M and with $L_M^{2/3}$, which also rises.
- r_T **falls**: L_F falls, lowering MPL_T ; P_F is unchanged.

Numerical check (optional but informative): new equilibrium solves

$$3 \cdot \frac{2}{3} \cdot 4 L_M^{-1/3} = 5 (90 - L_M)^{-1/2} \Rightarrow 8 L_M^{-1/3} = 5 (90 - L_M)^{-1/2}.$$

Solution:

$$\boxed{L'_M \approx 82.592, \quad L'_F \approx 7.408}, \quad \boxed{w' = 8 L'^{-1/3}_M \approx 1.837}.$$

Specific-factor returns:

$$r'_K = P'_M \cdot \frac{\partial Q_M}{\partial K} = 3 \cdot \frac{1}{48} L_M'^{2/3} = \frac{1}{16} L_M'^{2/3} \Rightarrow \boxed{r'_K \approx 1.185},$$

$$r'_T = \frac{1}{20} \sqrt{L'_F} \Rightarrow \boxed{r'_T \approx 0.136}.$$

Percent changes:

$$\Delta P_M = +50\%, \quad \Delta w \approx +44.9\% (< 50\%), \quad \Delta r_K \approx +60.8\%, \quad \Delta r_T \approx -30.9\%.$$

Interpretation: Capital owners gain the most; landowners lose; workers' nominal wage rises but less than P_M .

5. Labor's real wage and the "ambiguity."

Initial real wages:

$$\frac{w}{P_M} \approx \frac{1.268}{2} = 0.634, \quad \frac{w}{P_F} \approx \frac{1.268}{1} = 1.268.$$

After the shock:

$$\frac{w'}{P'_M} \approx \frac{1.837}{3} = 0.612 \quad (\text{falls}), \quad \frac{w'}{P_F} \approx \frac{1.837}{1} = 1.837 \quad (\text{rises}).$$

Conclusion: Labor's purchasing power falls in terms of M but rises in terms of F . Without a specified utility function (or consumption basket), the net welfare effect for workers is *ambiguous*, consistent with the core prediction of the specific-factors model in Krugman–Obstfeld–Melitz (Ch. 4).